

ActInf GuestStream 098.1 ~ Active Inference and Human-Computer Interaction

GuestStream | Active Inference, Human-Computer Interaction | 1:34:22

Hello, welcome. This is Active Inference Guest Stream number 98.1 on February 10th, 2025.

We will be discussing Active Inference in Human-Computer Interaction with the authors of this very interesting paper.

So, thank you all for joining and a pass to the authors for their introduction and presentation.

Thanks again, looking forward to it.

I'm Roderick Murray-Smith, I'm a Professor at the University of Glasgow and the PI of the European Research Council project that's funded this work, the DEFI project.

John?

I'm John Williamson, I'm a Senior Lecturer here at the University of Glasgow and also one of the investigators on this project.

And I'm Sebastian Stein, I'm a Research Fellow here in Glasgow and also one of the investigators on the project.

Okay, so we published on Archive a review paper on Active Inference and Human-Computer Interaction and then the friendly team at the Active Inference Institute said,

Oh, would you like to give a talk on this? So we're very honoured to be included in the Institute's presentation.

So we're going to give an overview of it, but if you want to see things in more detail, then at the bottom of the screen you can see a link to the archive publication.

Okay, let's get going.

So, we'll cover a bit of motivation for the work, describe Active Inference and HCI, what the core elements are.

We'll give an example of a particular case study in Ordinal Selection, which Sebastian will present.

And that's work that we presented in Oxford at the Active Inference Workshop, where we thought it would be useful to have some concrete examples as part of the paper,

as part of the discussion here. And then some of the challenges for applying this work in the context of human-computer interaction.

Oops. So, a little bit of background about where this is all based.

We started a year ago the five-year project called DEFI, Designing Interaction Freedom with Active Inference.

And the idea was we wanted to look at how we could better build the flexible interfaces of the future that would use richer sensing,

machine learning, machine learning in normal ways. And that has three main elements.

Active inference is going to be the framework for building things.

We wanted to use an approach called Optimal Mechanism Design to build the interaction mechanisms that could combine rich sensing, machine learning tools,

and embed content into computational structures. And we wanted these mechanisms to also have shared autonomy mechanisms so that the amount of freedom the computer would have could be adapted.

So, what we'll be talking about today is the first part of this activity on active inference.

So, there's a bit of background for the motivation. You can think about, you know, how have humans gone

about controlling the world around them?

So, you know, the very first tools would just be simple rocks and things where humans could use their muscle power to pick up a rock and do something more than with their bare hands.

As time went on, we had tools which would give us some more feedback about the world, that the actual design of a chisel might make it easier to feel the texture of the wood that you were working with.

And then further development got us to take external power and modulate that. So that could be wind power or steam power.

There's some external source of power which we could use to adapt our tools.

But recent years have moved towards the insertion of external computational power.

And you can see the framework that we're working with here. And the grey boxes in this figure are where computational power has been inserted.

So the human can act on an interface, they can perceive the interface, and that computer that they're interacting with can also control the world.

So this could be a smart car, and it could be tracking the road with its own cameras and augmenting the control to the steering wheel.

But it can also be looking at the human and trying to infer their attention. It could be augmenting the display.

So we have a closed loop interaction with the world, and the tool has got external computational power, algorithms, artificial intelligence.

So the argument would be that for less effort, the human could achieve the same or more performance.

And that's kind of a summary of what the AI was actually standing for artificial intelligence rather than active inference.

You can take one of these old Don Norman graphs where you're plotting the amount of effort you have to put in against the performance.

And usually the lower line would be human alone.

So usually you have to put a fair amount of effort before you get any sort of performance out.

I think artificial intelligence or algorithmic support can lower that threshold so you can get something for far less.

But it's always worth having a think about where that gap is being filled from.

Is it an earlier version of you that's somehow been learned or is it other people like you according to some measure?

Or is it more malevolent input from whichever service you're using which is trying to nudge you the way they want you to be?

So anyway, it's worth thinking about that background to tool design.

We're saying that humans are now starting to need to use artificially intelligent tools or algorithmic tools.

So these are going to be the most complex things that we've used to date.

And the tools are no longer going to be passive extensions of ourselves.

They're going to be active partners where they're starting to take on certain tasks and interaction becomes more of a dance than a command and control one.

So active inference appears highly relevant to this element of active tools.

Also, as part of the context, we think there's a bit of a crisis in HCI.

So you've got key researchers like Kornbeck and Roles who are highlighting that we have concepts of interaction that talk about design,

but we fail to produce theories and concepts that have high determinacy and adequate scope.

So they can't really do anything in detail and they're not or they're not broad enough if they can do something in detail.

We haven't in HCI been seeing the sort of ratcheting up performance that you see in subjects like machine learning and computer vision,

where people are sharing software and building on each other's work.

And there are key questions about agency, which is core to understanding how we are going to be relating to algorithmic support.

But how do we go about measuring that? How do we go about checking whether our agency is being taken away from us by computers?

So given that active inference is a closed loop computational theory for modelling agent behaviour, it seems very relevant to discussing these questions.

So you've got this elegant unifying theory.

It allows us to incorporate machine learning, Bayesian inferencing, probabilistic programming and dynamic systems.

We can work from low level physical interactions to high level reasoning decision tasks.

So it seems like it's potentially going to be something that could support the next generation of human computer systems.

So the elements that are relevant for that is in human computer interaction, you've got a lot of uncertainty.

Humans are very variable. Their context is complex and variable.

So we need to have models that can cope with uncertainty.

Active inference is completely probabilistic and views agents as entities performing Bayesian inference.

So that's quite appropriate.

Humans are also intelligent agents. They're inherently predictive.

They're trying to understand the world around them and act in a way that is based on predictive models.

Active inference supports this sort of analysis.

So we have a predictive element.

And we also want to balance the exploitation of the knowledge that we have with exploration to try and reduce future uncertainty.

And that's something which active inference gives us support on as well, unifying the reasoning about perception and reasoning about actions in a single framework.

So let's get into active inference and HCI.

So your traditional active inference framework was often set in a more natural context rather than a design context.

So you might have an intelligent agent.

In this case, we have our human user, but it could be any other biological agent.

And it's interacting with an environment.

And it has action states that it can sense the environment and make predictions about what it's going to sense given its actions on that environment.

So with active inference in a human-computer interaction context, there are several major configurations here in M3.

You've got an active inference model of the human interacting with some system.

It could be anything.

So when we're saying system, we're talking about the computer and its interface.

You could have the computer and its interface represented by an active inference model interacting with a user who's unknown.

You could have the active inference model transducing the information between the user and the computer.

So we can split things up into offline and online use.

So the description approach is saying we've got a model of a user operating under active inference principles to reason about how the user is going to behave.

You could have the system as just its environment and you could use that at design time to simulate user behaviour and optimise your designs.

Or you could have it at evaluation time to interpret observed interaction behaviours.

Mutual interaction is where we're modelling both the human agent and the computer agent.

So we're looking at a model of the whole interaction loop and we have two interaction agents working together.

So here you see one agent has become the environment for the other.

So the human motor states are then perceived by the computer sensor states and the computer's action states could be changing its display, making noises, sending vibrations, moving a robot arm.

These are perceived by the perception states of the user.

So the major change from the usual presentation is we're designing the environment of one agent with another agent.

So the user is part of the environment.

OK.

OK.

In terms of online options, we've got a user interacting directly with the active inference agent.

This is a constructive approach.

So the user is part of the environment, but might not be explicitly modelled in terms of theory of mind.

We'll get to that in a second.

And the other one is the transduction one.

So you can have an active inference agent lying between a user and an existing system, so it's mediating their interaction.

And neither of them are explicitly assumed to be active inference agents.

So the user and the system jointly form the environment of the mediating agent.

And all of the models that we've been discussing so far could be augmented by explicitly incorporating an active inference model of the partner within their generative form.

or forward model.

So we've called this a reflective approach.

You're concentrating an active inference system that incorporates an active inference model of a user.

So you see the embedding of another active inference model within its system predictions.

And this ties into work that's been going on in other areas of work in theory of mind in terms of levels of mental modelling.

where you could basically have different levels of higher order cognitive modelling of mutual theories of mind between humans and agents.

So we're now going to go back to the transduction work that we mentioned here and have a specific use case.

So Sebastian, do you want to take a look?

Yeah, so this is a use case where we look at what's called noisy ordinal selection.

So in this case, we were looking at using active inference as a transducer as Ron suggested, and as one of the options that we might want to use active inference for in human computer interaction.

And this is work that we presented at the active inference workshop last year in Oxford.

So here we're looking at a rather abstract human computer interaction setting, which you can think of as a two player game where one player, in this case, the user would think of a number.

And the other player, in this case, the interface attempts to guess that number and presents one number in each interaction time step back to the user where the user could then give feedback in the form of saying that the number that they're thinking of is either higher or lower than the number presented.

But that feedback is then corrupted by some amount of noise where the feedback below might be flipped into the feedback higher and vice versa, potentially with different rates.

And that's a sort of scenario that's rather relevant for sort of naturally ordered symbols like alphabets or sort of imposed orderings like we have in things like drop down menus.

And the sort of channel characteristics of having very noisy feedback is quite commonly seen in interfaces like brain computer interfaces or other assistive technologies like electromyography, where we usually have rather low classification rates and have to deal with high levels of noise.

But we have the possibility to have rather noise free sort of feedback in the form of a display back to the user.

So here's one example of how an interaction might unfold over time.

So time is on the x axis here and the different targets presented to the user are indicated by the agents trace and presented by the blue line here.

In this case, the user is thinking of the number four.

And then the first time step all the way on the left, the agent or the interface is presenting to the user the number 16.

The user feedback is indicated by the triangles pointing up or down indicating that the feedback received by the interface was either lower and in the case of this first time step or higher.

And in the case of the second time step.

What you see here is that I think in terms of five it is where the feedback given by the user is the first corrupted feedback, at which point the agent flips effectively its exploration area to sort of higher.

The higher numbers and we will see later why that is in particular here the agents flips its belief to be one

where the user is responding with probably high error rates.

We see that over time the system recovers and the agent can actually correctly infer that the user's target is at or near the true target number four.

Next slide.

Okay.

Okay.

So there are some classic information theoretic approaches that we could apply here and which have been applied in previous work.

So we can see that in the same way, but they are only optimal under a set of strong assumptions.

For example, the horse times posterior matching algorithm is only applicable if we assume that the noise is actually i.i.d. fixed and known in advance.

And that the symbol set is arbitrarily large as well as the feedback channel being noise stream.

So we were interested in seeing whether active inference can help relax those assumptions.

In order to do that, we translated the problem.

Yeah, if we can explain.

We translated the problem into an active inference formulation where we represent the user's intended target as well as the error rates as the

latent state of the system that we would like to infer.

And the observations that the agent can make about the system, its reasoning about the user are these binary feedback symbols, which correspond to the number being lower or higher than what the agent has displayed.

And the agent's action space is effectively the set of symbols that the user might want to select.

In order to apply active inference, we need probabilistic forward models of the system the agent interacts with, which in this case is the state transitions of this latent state as a function of the agent's action in the previous state.

So that's kind of answering the question format, the user's target and the noise vary over time.

And we need a probabilistic forward model of how the user feedback is generated.

So answering the question, what symbol would I observe if the user intended to communicate that the target they're thinking of is higher.

And so in order to select the next action, we need to inherently trade off between learning about the user's intent and learning about the environment characteristics of the error rates of the channel.

So commonly we use Gaussian distributions to represent the belief about the latent state in active inference agents, but Gaussian distributions are not particularly helpful here.

That is because a sequence of user interactions is usually compatible with distant and mutually exclusive beliefs.

So if we take an example where the agent displayed the number 20 and the feedback received was that the user's number is lower than that.

And that is compatible with the user's target actually being lower and the channel having a rather low error rate.

But it's also compatible with a scenario where the user's target is actually higher and the channel has a

higher error rate.

So in order to represent a potentially large number of these modes in our belief distribution, we decided to implement a particle filter.

We have the belief distribution is then a weighted combination of point masses in this space of latent states.

But the K-divergence can still be computed rather efficiently and quickly just by using the multinomial distribution represented by the weights in this space.

So going back to the example that we saw in the beginning, I want to now look at how the agent reasons about the latent state, in particular the error rates and the user target.

And we're going to first look at the error rate, which in this case we assume to be symmetric.

So it's equally likely to flip a signal lower into one that's suggesting that the target is higher and vice versa.

And here you can see around the time step where the first errors occur, the agent belief about the error rate is sort of changing from a lot of mass accumulated near the lower error rates to mass in the belief distribution.

And here you can see that the error rate is aggregating around high error rates.

So I should have said before, the color in this case represents the probability density within sort of local regions around these sort of discretized levels of error rate.

And the true error rate in this case is 20% as indicated by the red dashed line.

And so over time we see that the agents belief about the error rate converges to somewhere close to the true error rate of the system.

Next slide.

Now looking at the marginal distribution about the user target, we see that initially the agent has rather broadly spread out density, meaning that there's relatively low evidence for specific targets over the first few time steps.

But as the agent learns more and more about the error rate with which the user is communicating, it's unable to identify the target correctly towards interaction step 35 and onwards.

So we did quite a comprehensive set of evaluation experiments which are detailed in the paper, but just to highlight a few here, I want to present first what we've done in terms of simulating experiments and then present some of the key results.

So we're always assuming that the channel statistics are symmetric in this case, just to simplify the analysis.

And we're looking at scenarios where the user could select between one out of 32 different targets.

And we repeated simulations where the simulated user actually imagines each of the 32 different targets 10 times and simulated for 500 interaction time steps where after each, every 100 time steps, we reset the distribution over the user targets to be uniform, which is simulating the behavior of a five successive interaction episodes.

So one of the main findings is that even in the absence of knowing about the error rate of the communication channel, the active inference agent can make decisions as well or better than the Horstein algorithm that I've alluded to earlier.

After the first 300 times, when the agent has learned enough about the error rate to compensate.

And we've also done some experiments indicated on the right hand side of this slide where we looked at the error rate actually changing over time.

And again, with the heat map displayed here, you can see that the agent's belief can track the changes in the

true error rate of the user.

So just to summarize, we presented a concrete example for using an active inference agent to do reliable one of end selection,

and showed its effectiveness in inferring both the control polarity and working with non-stationary channel statistics.

And we can bring in additional assumptions and further domain knowledge in a setup that's more specific than this one.

Thank you.

Okay, so let's talk a bit about core elements of active inference in HCI.

So the key aspects of an active inference agent are listed below.

We've already discussed the idea of the agents and what entities are going to be modeled and implemented as active inference agents, the users, the systems, both, neither.

We need to look at the role of the environment.

Is it just a transmission medium between the agents?

Is the user using the system as an instrument to act on it?

Or are they both fighting environmental flux operatively?

We want to tie in some discussion about Markov blankets.

How do we separate agents from their environment?

Work out the actions that each agent's capable of and be able to define boundaries between the user and the system.

Forward models are obviously key to active inference.

So we have to be able to create a generative forward model.

And that's often going to be an implementation step that's very challenging.

And because we have to be able to predict future states, synthesize the observations that we would see.

We're going to look at how we predict and reason about the next best action because we need to use the forward predictive models to estimate future states and predicate actions on the estimates.

We have to think about the preference pile prior structure because the agents goals or desires are going to be defined as a distribution rather than as a single reference value or control law.

So we're going to have to think about how we can shape these priors to align with the goal of interaction design.

So it's this change from trying to describe the natural world to trying to design and create a system or a world that's going to be better for the humans involved.

And then there's all the practical implementation questions around building and training the forward model, implementing your rollout policy, getting your inference mechanisms to work and perform Bayesian updates and the challenges associated with that.

So let's start with the environment. Here's a sort of cartoon of the basic ways in which the environment can play a role.

You might have the user in the system who is directly engaging with the environment. Can they do this, manage the variability in the environment cooperatively?

So an example of that could be that you might be trying to use a smart speaker and the users try to operate

that to change the mood and observable behavior of a party in their flood.

But the user can also interact directly with others in this party and they can bring their observations of that together to pick the next best track.

Similarly, a central heating system might be might have an application for controlling the temperature in the building, but the user can also make changes like opening or closing windows.

We might have a transmission approach so that we were just describing the environment in terms of what to what extent it affects the communication between the user and the system.

So if you have an environment with lots of background noise, then any audio display by the system may be difficult for the user to hear.

Similarly, bright sunlight on the screen might make the screen visible to the user.

So we need to understand the impact of the environment on the communication between the user and the system.

We may have the user applying the system as some sort of intermediary instrument to act on or better sense the environment.

So they could be transducing user unobservable information from the environment.

Or it could be acting on the environment at the behest of the user.

We haven't discussed multi-agent environments, but they would be a natural follow-on where we start to have multiple explicitly modeled agents acting together.

Okay.

An interesting question in HCI is where the boundaries are.

And obviously the active inference community is focused quite a lot on the notion of Markov blankets.

But HCI has often struggled a bit with this.

So the very name human computer interaction suggests that we're assuming there's a clear boundary between a human and the computer.

And people might often potentially naively associate that with Clark's skin bag around the physical limitation of the human.

And we know that if you're going to understand the system, then finding the boundaries between that system and its environment seems to be essential for analysis of it.

But these, the Markov blankets that describe the limits of one agent may change dynamically as we give and take control.

That could be driven by technology.

So we might give a user a tool or an assistive technology and we can transform what they can act and sense.

So that if you have somebody wearing augmented reality glasses or virtual reality, then we can change how they're interacting with the world.

We can let them see things that would otherwise be invisible.

And we may be able to give people the ability to act in the world in a different way.

So modern tools will have context sensitive algorithmic support from algorithms from AI.

And these could augment the user cognitively or it might relieve them the need to attend to or check on mundane tasks.

And so that could be seen as taking or augmenting agency depending on how it's designed.

But basically you can imagine that as we're taking over tasks from the human that the states which previously were being controlled by them are now being handed over to the agent to work on.

So we're seeing an adaptation of the Markov blanket depending on how much support we're giving the human.

So we've been evolving human tools from rocks to AI as we mentioned earlier.

We're getting this trajectory of increasingly powerful effort leverage, doing things better, doing more new things.

And so that means that we're increasing our individual collective intelligence.

We're extending our natural selves with our tools.

And although we started off with crude functional tools like hammers and axes, we've moved to tools which have controllers in them like thermostats and governor technologies based on a controller approach.

And then each time you're giving some of your agency away for understood outcomes and the tools can take on more and more of our tasks.

But the question is, are we at a point where we're potentially giving up really important bits of ourselves?

Okay, so arguments have been made in the active inference community that to maintain the integrity of the agent, the blanket needs to change.

And there's several papers making arguments about that, that you need to adjust yourself as your environment changes in order to keep your internal states stable.

And we're looking at the basic approach that we were showing earlier, we have the Markov blanket with its sensory and actuation nodes, internal and external states.

But I think one of the interesting things that we could be pulling in from machine learning is the notion of attention.

So that at different points in the interaction cycle, the human may be actually only paying attention to a very limited subset of the sensory states.

So we have an effectively different Markov blanket, even though the human could potentially perceive things on the states that they're ignoring.

And this can of course be structured into a hierarchical setup.

So this use of attention means that we could be tracking, you know, with our closed loop models, we could be going through interaction processes and looking at what the human pays attention to using the standard Bayesian methods to infer Markov blankets at different points in time and see what is the human ignoring at certain time points.

And what does that mean for system design?

Other research within HCI has been raising similar questions, although not from an active inference perspective.

So there was a paper by Satnian and Jones at MIT, that one's a computer scientist in an HCI context, the others an anthropologist, and they were looking at intelligence as agency.

And looking at the co-fused agents and looking at the co-fused agents and looking at the line of control shifting between them.

And where they say composed, we would say acting on the environment.

And when they say interpret, we would think of that as sensing the environment.

And a lot of their interest was in different contexts.

And they were arguing that in certain cultures, one is more appropriate than the other, but it's not consistent across cultures.

They were also speculating about what happens as your AI is taking over more and more control over the state of things.

As you can see in the bottom right corner, this sort of Pac-Man figure where the green is gradually eating up the blue of the human.

And so you can think about that relating to Markov blankets and saying, well, you know, which states is it important for the human to continue to have control over them?

Okay. We mentioned forward models.

In active inference in a human computer interaction context, we're going to need to have forward models of human perception.

We're going to have to forward models of the computer sensing process.

We're also going to need to be able to predict the human motor control system and what it's capable of, and the reliability of the computer displays given the environment.

An interesting observation is obviously that active inference focuses on forward models and it doesn't need explicit inverse models,

which is a difference from other contexts.

Okay. So we want to look a bit at prediction.

So active inference agents aren't reacting directly to the sensations, they're reacting to their beliefs about future states of the world,

and the observations are there to correct the estimates of these beliefs.

So we're trying to minimize our future surprise.

And you've got the standard approaches of expected free energy and potentially breaking that down as information gain and the pragmatic value.

And the interesting question is how humans are going to experience this.

So if a system is acting to gain information about them, they may experience it as a proactive system, giving them a personalized way to interact, or it may be that they find that the system is moving into the uncanny valley,

where they're feeling distinctly creepy as it tries to pick their brains about what sort of a user they are.

So it's going to be interesting to see how the implications of curiosity in the information gain term are perceived by users when they start to interact with the system.

But we also think that the whole approach of using surprise minimization as a guiding principle in interaction design is going to give us a normal perspective on the design process

and may lead to some surprising outcomes in terms of final system design.

Okay, preference priors are obviously another key element of active inference.

And researchers initially have highlighted that most concepts in interaction don't tell us much about how intentions are formed or affected by interaction.

And it's interesting to think about how active inference manages this, how the preference priors are going to be designed,

because obviously our systems behaviour is going to be a function of the preference priors that we give the system and what we assume about the user.

And there's more recent work in active inference about adaptation of preference priors during interaction, and it's going to be interesting to think about how that models, but that seems like an area that's still quite open for development.

And we'd be curious to hear how the community's evolving in that direction.

Okay.

And there's, you know, a lot of people in human-computing interaction have been used to using objective reward functions in reinforcement learning,

and so there's interest in contrasting the active inference approach, misdistribution of preference priors compared to classical control laws.

So we'll see all those.

Okay.

Challenges.

It's not going to be easy.

Obviously, there are huge computational challenges in active inference with the sorts of complexity of system we're looking at.

We're going to have all the complexity of action rule-ups.

We're going to have to balance the performance of the final system with the real-time requirements of online applications.

And it's going to be interesting to see how much we can continue to use sampling and research, and how much we have to move towards amortized algorithms where we're learning a mapping from offline behavior.

Even if we can't get real-time work going, we do still think that offline analysis of these mutually interacting active inference models can be interesting for the design side.

We're essentially creating digital twins of our human and system and observing how they interact with each other.

Another challenge we feel at the moment is that the software infrastructure for active inference is still lagging a long way behind,

for example, machine learning.

So having easy-to-share, standardized software libraries that people are familiar with,

and having the correct computational abstractions to build and reason about interfaces is not there yet.

So we need to develop workflows where we can actually apply active inference to interaction in a predictable way.

As we just mentioned, preference modeling is a new perspective for people in human-computer interaction, in terms of using distributions.

So we've got these flexible frameworks, but they're unfamiliar to designers,

and the consequences of changes to these distributions will have significant effects.

So how do we work with that?

How does it manage the uncertainty in the heterogeneity of human behavior?

And in general, with interfaces representing uncertainty,

has the potential to make interfaces more robust because we know that we have that uncertainty there.

Ignoring it is not an option.

But if we create problematic interfaces, users may struggle to understand what's going on.

So creating the right metaphors, mapping the uncertainty into usable feedback for the users is going to be a major challenge.

Prediction is obviously key for active inference, and that can potentially mitigate latency in the interaction loop.

But that predictive aspect of it could be unsettling for users.

It may make the interface feel like it's in this uncanny valley where it's sort of human but not quite.

We've been speculating as if we have a pair of active inference agents,

they're going to be trying to push the joint system into a mutually predictable region.

So that even if their predictive and generative models are not yet completely capable over the entire range,

they're going to be, if active inference is the right framework to think about things,

then the two agents are going to be pushing each other into a mutually predictable space.

So that could mitigate some of the challenges that we've just been outlining.

In terms of preference, we're going to have to develop how we elicitate and validate preference priors for design purposes,

how we can get generalized preferences from individual user preferences,

but potentially creating hierarchies of preferences.

Okay.

Lots of the criticism of an active inference approach from the HCI community would be similar to other computational simulation approaches.

You've got the cost and complexity of developing models,

the general difficulty of creating models which can represent the cognitive and perceptual complexity of humans,

and especially the sensitivity of human behavior to the fine details of context.

But, you know, there's something that's been around, a challenge that's been around for a long time,

and Thomas Sheridan, one of the people who's been working in this field since the sixties,

has made the point that, you know, actually the process of modeling forces you to really think hard about what you're considering

and to ask and answer questions about what the essential features of the structure and function are.

So, in terms of changing the approach to an HCI to doing science, forcing yourself to create good generative models

is an important part of thinking about and understanding the problem.

So, the modeling that's required by active inference requires you to think about what are really important features,

and if you can build generative models that validate against user behavior,

you've got an acid test of whether you really understand that interactive system.

But, of course, any adaptive intelligent system is going to have to deal with co-adaptation of users, and you're going to have to account for recursive theories of mind.

So, we're facing really challenging things because even humans sometimes struggle with predicting what they're,

the people they're interacting with are actually thinking and what's going on in their heads.

Okay, so active inference faces the challenges, but you've got that inherent capacity to model those and then you've got the user and the computer in an interaction loop, and then try to approach the problems computationally.

Okay, so, in general we're seeing active inference as an example of the broader area of simulation intelligence in HCI.

If you haven't heard of the term simulation intelligence, there's a very good archive paper on it, it's about 80 pages long,

but it goes through how you can bring different types of simulation model into understanding the world and making better models of it.

And we see active inference as very much an example of this.

You've got a number of different fields coming together which are letting you build a computational human-computer interaction approach,

and that can include interdisciplinary approaches.

You've got probabilistic programming and predictive models.

You've got physics and from machine learning.

You need to take dynamics and control systems into account,

and you've got the physiology and cognitive science of the human side of it.

So, you're simulating the whole interaction loop from perception, cognition, motor control, sensors, computation, display,

and you're doing that in a way that's computationally implementable and testable.

And we had some previous work based on non-active inference approaches, which was published at the WIST conference over a couple of years.

And this is us simulating human motor control and perceptual systems.

So, there's a simulated ocular model, simulated muscle models,

and then that was using reinforcement learning to perform certain tasks.

And it was the user-in-a-box model that you could simulate.

We would like to extend that work to include active inference for the future.

Okay.

So, one of the things that we're arguing is an important part of this framework is that having active inference models of both the human and the system lets us model the intertwined action of human and system.

And it's avoiding the problems that were seen when people tried to bring classical control models in, where you could exchange building blocks in and out, assuming nothing else would change.

Because already in the 1960s, Macriar's colleagues, with their crossover model, demonstrated that human pilots would adapt their behaviour to shape the properties of the closed-loop, so that if the aircraft properties changed, they would become different pilots, they would behave differently, so that the closed-loop system had a consistent shape.

And Holnagel's joint cognitive systems approach highlighted the need to examine the joint system rather than individual blocks.

And so he was viewing the human as an implicit model of the process they were controlling.

So we can't understand interaction either from the user or the technology in isolation.

You have to have this joint model.

And we think that this sort of active inference, mutual interaction models are going to be required to actually model concepts like agency or engagement and interaction.

So let's have a look at our outlook.

We moved into this area because generative modelling is at the core of active inference, and the massive advances that have been made recently in generative machine learning and make it possible to do the sort of analysis of the human perceptual system that would have been impossible even ten years ago.

So the whole general...

Before we started working with active inference, we were using things like perceptual control theory and systems, so we were taking a closed-loop approach.

We were looking at empowerment approaches, you know, Polanyi's work, where you're modelling the channel capacity from the actuation to the sensing.

But these control-based approaches will only become realistic if you can model human learning perception.

And that's what machine learning has provided us with in recent years.

Active inference is flexible preference, and continuously adaptive behaviour we think is going to be well-placed to support AI tool use.

And we've highlighted the implications for HCI theory, so having this whole loop model gives us the potential to model agency and engagement, although there are going to be challenges in getting there.

And active inference gives us this probabilistic predicted unified approach to closed loop modelling that we think could be quite a clean representation of the problems.

We've highlighted that there are major computational challenges in getting things going, especially in real-time.

If it's a good theory of behaviour but not yet practical for real-time use, could it be used as a better user in a box model to help us in the design process offline?

And there are going to be interesting questions about when to amortise the computation into some hard-coded non-linear network.

But of course, you know, we've given you one example during this talk of an interactive application,

but we want to branch out within our project and test it in a wider range of interactive applications because that's how we think we're going to hit prop this.

So that's us hitting the end.

And I'm going to just put up one thing that we had good feedback from reviewers about was our one-page visualisation of active inference.

And you can see that developed in the actual archive paper itself.

But some people thought that was a nice way to present it.

So we'll leave that up while you ask questions.

Okay.

Thank you.

Awesome.

All right, Aaron, want to begin with some reflections or questions?

Yeah.

First of all, thanks very much for your talk.

It was really interesting.

I'd also like to echo what the reviewers said.

This is a fantastic introduction to active inference if you do not know how it works.

Really clear figures, clear explanations.

If you're about to start reading, say, the PAR textbook, I would say have a look at this first and then go to the textbook.

It's a really just nice primer.

So thank you for that.

Like genuinely, education and communication of how active inference works is really hard.

So it's really nice to see good examples of how to communicate those ideas very clearly.

In terms of my questions, I have loads.

Daniel, if there are questions from the chat, definitely interject and put those in as well, or I will just monopolise all of the time there.

But to start with, I think what I wanted to start with is in terms of the alternatives to active inference that you experienced before,

what were their key limitations that you think active inference is completely overriding?

and what are the limitations where active inference is also going to struggle?

So, as I may, I did mention the things like perceptual control theory, you know,

where you're saying, you know, behaviours, the control of perception.

So there's a lot of the general ethos of active inference already in there, but they were not.

So behaviour, perceptual control theory wasn't explicitly using a probabilistic approach.

And it had, you know, standards reward function.

So there are lots of, you know, lots of different areas where people have got elements of it.

So the active or the empowerment approach where you're modelling the channel capacity from your

actuators to your sensors,

that's essentially saying how much can I put into the world and predict what's going to come.

So that's relating to generative models.

So there are similar points there, but active inference has a nice coherent framework.

We're still trying to understand ourselves what the relative merits of, you know,

the energy and expected free energy and the energy of the future and so on.

All the variants that could come in there as an actual cost function are.

So that it feels like there's something interesting there with the preference priors giving you more flexibility.

But we don't, you know, we need to see that making an improvement on a concrete case and understand the difficulty of how to specify those.

Because when you're not just trying to describe a natural system, but actually create and design a system that's never been made before,

you have different possibilities and it's potentially taking active inference out of its comfort zone in terms of not just describing natural things,

but saying, is this a good design approach?

So I'm not sure I've answered your question, but if John or Sebastian have other ideas to answer it better.

Yeah, we've done quite a lot of work on Bayesian models in HCI, but that's usually about how we model the absorption of information for the environment solves the perception problem.

And one of the nice things about active inference is that also includes the acting part of its brain box.

So it really adds to the whole interaction loop can be unified in that model.

I think that's one of the big advantages.

I think also the predictive aspect.

So not always being about modeling what is happening now.

That is really important in human interactions.

You have to deal with latency and the fact that people are also making mutual predictions.

Most of the models don't explicitly account for that.

And I think that's something which active inference does very well.

And there are some approaches like computational rationality where people focus on every area will have its own focal points.

And computational rationality, there's a lot of focus on what are the cognitive bounds or the physical bounds of the human.

How does the environment have constraints?

And these would all be represented by the generative models.

And active inference also, the community focuses more on, say, the Markov blanket aspect, which is maybe underplayed in other areas.

And so that's interesting from an analytic perspective, if you're trying to understand things of systems and say, well, what is the system?

And how does that change over time?

And how should we infer that?

So, again, how do we, we can take the general ideas of the Markov blanket, but actually making that work reliably doing the statistical analysis to be able to actually say, okay, no, we can actually infer this problem.

But actually, how much data do we need to do that?

And how reliable is it going to be seems to still be an open question.

But that's an aspect that's not been a focal point of other areas.

So I'll see anything more.

Okay.

So.

Fantastic.

Next question.

But maybe we could also talk about the vignettes at the end of the paper in the appendix, because I think they were very instructive.

But I'll ask this question, and then maybe you could talk about the vignettes in relation to this question, which is, what do you see as a difference between computation and cognition, when it comes to these sorts of computer assistance for humans to use?

Not sure I understand the question, but if any of you understand better, or think you understand it, then have a go.

So.

I'm not sure I understand it.

No, sorry.

No, no, sorry.

Maybe I'll come for an example.

So let's say I have a calculator that as a tool can help me compute.

But I wouldn't necessarily say it's taking away, it's not shifting that line of control that you spoke about before.

Okay.

Right.

Okay.

Whereas we could extend the sort of this create a spectrum, say, where there's a tool where let's say it's a large language model power chat bot, where I could ask it to tell me something or do something for me.

And that's going to, maybe that's the question a bit.

Yeah, take the example of the smart car.

So, you know, you might have your smart car in a full manual mode, and then you're having to pay attention to everything in the world.

And, you know, you're having to make sure that you stay in your lane, that you keep your speeds, you know, appropriate for the other cars around you and safe, and so on.

And you have to be aware of the how slippery the road is, icy or wet or dry or sandy.

And if you move into a semi-autonomous mode, where the system starts to do lane following, you are now relieved of the, you know, the computer can do the computation of tracking the white lines in the road and making sure that you're in a safe distance from both sides and safe distance from the car in front.

So you're relieved of having to monitor those states in the world.

And you can potentially use that spare capacity to do something else and think a bit more about where you're actually trying to go or whether to change your route and so on.

So that would be an example of giving up one bit of agency so that you could focus on others.

We had the smart speaker example that could be combined with the car.

So there the smart speaker could make decisions about what to play.

Now you could have something like a control, a dial where you could gradually move through different genres of content and have much more control of what exactly was going to come up next.

But if, for example, you're trying to control your car while also picking music, you maybe need to give more autonomy to the music player and just let it play something roughly in your direction rather than trying to micromanage it because you've got to keep your eyes on the road.

So it's this divided attention, you know, what are you paying attention to at what time and what's a reasonable amount of attention to pay that.

It would be unsafe to focus on the music system when you're supposed to be driving.

And so active inference could help in predicting what you're doing, what your loads are, what your limitations are.

And if you're spending too much time paying attention to the interface, it needs to take control and take that away from you so that you can focus on the road.

Does that make sense?

Yes, yes.

Next, I'm just going to go through my list.

It's sort of in the order as we spoke about some things and I'll also pull some stuff from the review paper as well.

But yeah, if you feel like there's anything else that you didn't get a chance to speak about that you'd like to put in, then by all means do so.

So my next question is on the uncanny valley aspect that you mentioned before.

I think one of the vignettes was talking about like some of the like I think referred to like the Seal Paro product.

Not necessarily by name, but this idea of a robot that can anticipate your emotions and your emotional state and try and behave in a way to make you feel better.

Right.

And that's that's a product that has historically, I think the designers of that said they started with a dog.

And the reason they, the dog was not successful as a robot product to help people feel better was because people have very strong prior beliefs about what a dog is, how it looks, how it feels, how it sounds, how it should behave.

And that immediately put it in the uncanny valley because it sort of looks like that, but it's not.

And then that difference between your observation and your priors, presumably gives you a very large free energy gap and that is uncomfortable.

So I think it's very interesting to see how if you have, if you sort of make your robot more predictive and able to handle uncertainty, how that would actually create an uncanny valley effect.

In my mind, it would actually reduce that because it's able to better match your priors if it has a model of you.
Does that make sense?

Yes, indeed.

I think it's, it's going to be obviously, as you, you know, if we get to the goal of, you know, fantastic model of everything, then, you know, then hopefully things would be good.

But the, as you're moving towards that goal, there's going to be mismatches where it's slightly wrong and it's whether people find those glitches on the way to, you know, so if you had somebody who's, willing to stick with it, no matter how creepy it was on the way, then fine.

But it may be that some of the early phases of that learning process make it, you know, it's got some elements of prediction, but it's creepy enough that you just don't want to touch it.

It feels like, maybe it wriggles the wrong way and you just go, Oh my God, what is this alien thing?

And I don't want to touch it.

And so it never gets the rest of the information it needs to improve his model.

And it's the same with in general with HCI.

So we were talking about this in Copenhagen a week or two ago.

And there was a question about, are designed agents compatible with the free energy principle?

Because, you know, they don't die and so on.

But any, any app which is not good is going to get binned and any product, any robot that is too creepy is going to be switched off and ignored.

So death for an agent is being ignored and being thrown in the bin essentially.

So it's trying to do things to avoid that drop in its ability to act in the world.

So it wants to, if it's a, if it's been designed to work well with the human, then it should, you know, the human's preferences should be key to it.

But it also wants to be able to continue to work with the human in order to help the human.

And if it does something which gets it binned, then it's never going to be able to improve the human.

So that's a very black hole in its predictive horizon that if it thinks there's a chance that you could be binned for behaving this way, that's going to be something else to avoid.

So I guess it's the question is, will we hit uncanny valleys when we're on the way to perfection, when we're still making errors and we haven't quite got the model?

Or is it just already very weird if, if humans, therefore humans seeing a computational object that seems to be getting them so well, will that already, you know, be something that throws people?

And they, they don't, you know, it's partly also how quickly can the human adapt their model of the world?

So there may be some humans who've never experienced a machine that could understand them the way this one does.

And that's already hard for them to adjust to.

So they just say, no, this is not a world I want to be in where, you know, machines behave like this.

So that's, I think we're going to see different things from different cultures, different communities, depending on your upbringing, what your expectations are, whether you can grasp a machine that seems to understand you in some ways better than you do yourself, because it may be able to measure aspects of your behaviour and compare that to huge numbers of other people.

And maybe it picks out some things that you're not aware of yet, which would be potentially scary for some people.

Absolutely.

No, there's definitely very strong ethical considerations going into this, especially considering what do you explicitly model and what do you implicitly model?

And as you alluded to there, the group priors and then going for specific priors on an individual user.

Certainly what comes to mind when you're talking about products and going extinct, the extinction pressure is one tool that did not have, to my knowledge, at least any predictive power, but was very much largely a rule based tool was the Microsoft Clippy Assistant when you were typing in Microsoft Word and it goes, oh, it looks like you're trying to write a letter, which I think universally everybody found not necessarily creepy, but just incredibly irritating.

So I think that's where I sort of see us moving away from is these very simplistic assistants with very limited perception.

We go to something that's more probabilistic, more sophisticated.

Oh, sorry.

Clippy was the team, you know, the original work there was the Bayesian reasoning team and Eric Horvitz and co were doing the original work.

But I think the problem there was it was taken away from the Bayesian reasoning team and given to a product team who just said, bugger that we're going to clip all these things and make it much more straightforward.

And then, you know, they didn't have the richness to actually cope with the complexity of the environment.

Yeah.

Okay.

Could I ask a question about the, in sort of the experiences your team has had in actually implementing active inference, the noisy ordinal task selection experiments as well.

So I know part of the challenges that you raised was sort of the not necessarily mature software toolboxes that we have for implementing active inference.

So things like Pymdp, RX and FIR, these are probably the most advanced implementations out there for active inference.

Do you mind if I ask what your team was using and where you see the biggest challenges from the software implementation side?

Yeah.

Yeah.

So we, we looked at Pymdp and we did use it in some preliminary experiments, but at the moment, or at least at the time we were looking at it, it was very much limited to discrete probability distributions.

And so as soon as you get into very high dimensional state spaces, you reach the challenge that you have a combinatorial explosion of discrete bins in that state space.

And so it would become very, very costly, very quickly.

And so we were looking at alternative approaches and we don't use Julia at all.

And so RX and FIR wasn't really something we were looking at in much detail.

And so we just decided to start implementing some of these things ourselves.

And because we'd been working with realistic inference tools before we thought we were sufficiently capable of doing that.

But then actually, you know, deciphering the active inference equations, the expected free energy and so on from the book was more challenging than we thought, I think.

And so in part, it was also a useful exercise to actually do the programming ourselves to see, you know, do things actually work the way they should?

And now we're implementing the right thing here.

Yeah, so the short answer is we use our own implementations, but very much with the idea that, you know, as we work through more and more specific examples, we end up with more robust implementations where we hopefully can, you know, slot in categorical distributions and swap them out for particle filters or just guessing distributions and different inference algorithms.

And always depending on, you know, what's what's demanded by the individual application that we want to build.

Daniel, I think you can probably speak more.

We've been surprised at how challenging it is to do with things that we think should be simple.

So, you know, so it's been taking us longer to get, you know, we're building a basic mouse model, expanding. We had control, more predictive control based models of mouse movement in previous papers before we started using active inference.

And we thought it would be relatively straightforward to take the active inference approach to that, but it's taken us longer to get things going there than we expected.

So, yeah, so implementing things in continuous space is interesting.

And as soon as you get something concrete, you start to go, okay, so what should the preference priors be on in this, you know, so is it the selection process?

Is it the movement process? Should we have hierarchies of them?

You know, so it becomes a, it makes you, it forces you to suddenly go, okay, there's a lot more complexity here than we thought initially.

Yeah, I must admit, most of my experiences with the discrete models and PYMDP specifically, but I think Daniel can probably speak to more of the RX and FIR side and the Julia side.

And I think they are pretty advanced ways of handling the continuous distributions, particularly for the drone flying examples that they have.

Daniel, I don't know if you want to add anything on that.

Yeah, I really resonated that a lot of the considerations of cognitive modeling, independent of whatever implementation approach are like what slice or what joints of nature do you really want to model.

So that was a cool constructivist approach that you took with building up that biophysical realism.

And then when it comes to implementing active inference, it's a, it's a great learning experience to actually try to implement in limited cases, different equations.

And then that helps like pull back the curtain and realize like, hmm, we're not really dealing with a collected data set.

And then training a generalized statistical model to interpolate something about the patterns of that data set,

but rather explicitly stating certain distributions that might have observability or non-observability or control.

And, and so it's like pulling out those islands or the icebergs that you want to model explicitly rather than it just, it's like etched away or revealed from where in another way of modeling, you could have just collected the salient data set and then interpolated, not needed to specify anything specific.

Just, just, we have traced data on experiences that were rated positively or negatively.

We trained a model and that's how we are continuing to classify these traces of data versus it's like, wow, there can be a lot of steps and a lot of degrees of model or freedom in determining what, which explicit distributions are relevant for that.

It's, it's, it's, it's a separate question than like, which data set is relevant for it.

But, but that's, it comes up all the time in, in modeling and the frontiers of the software development are also quickly moving, hopefully meeting up with these more model or challenge areas.

The only thing I'd add on to that is coming back to, I think something you said earlier Rod on.

It's an acid test, whether you can create an agent that is successful in these environments.

Part of that is also actually not even necessarily if, you know, you've implemented the model correctly, but just being able to specify a generative model for a problem.

It really does force you to think.

And I think that's one of the things that's been missing a lot from the more traditional sort of reinforcement learning and deep learning approaches where it is just get a bunch of data, chuck it at a neural network and hope for the best.

And you sort of sort of relying on the fact that, well, I don't know what a generative model for this is.

I don't have any model of causality.

I don't have any explicit model of a user.

I just throw some data and I get an output back out.

And I think it's really interesting from the active inference point of view of getting explainability from structure and being able to think about a problem.

So almost like reasoning from your armchair and saying, I'm going to think about this problem and say the world acts in a particular, the world behaves in a particular way, specify those causal chains, and then test that agent out in the real world afterwards.

And I think the, you know, obviously the key thing is if you're using your deep learning to model how a city works with traffic or how a hospital work or how a company works, you're usually doing that in order to do something else.

more efficiently.

And so as soon as you close the loop using these predictive models, you've changed the city, you've changed the hospital, and it's not going to, it's going to be moved to somewhere where you have no data before because you didn't have the ability to control it before.

So the, you know, the active inference, because it's a closed loop framework means it's going to then have to adjust to that new world.

And that's part of the basic framework, whereas a lot of the classical, you know, big data work was saying, Oh, all we need to do is get lots of data.

We can build a model, but they were forgetting that as soon as they used that, they moved the system that

they were trying to optimize to somewhere it had never been before.

And their model was then useless.

So I think that's the interesting, but the challenge is, okay, we close the loop, but you know, like control theory spent decades looking at stability of closed loop systems.

And, you know, we're going to have to revisit a lot of that in active inference work is, yes, we're trying to change things, but, you know, can we guarantee that this, that our hospital is going to be stable, our city is going to be stable if we put active entrance loops around it.

And I don't think we can guarantee that yet.

Can we?

Lots of interesting questions in the chat.

Just read a few in this last section.

So Frasier Patterson wrote some questions, hopefully that we will hear more about in the coming years.

So let us continue on some other questions.

Thank you though, Frasier.

Jeff wrote, how can incorporating active inference in HCI assist user trust?

Well, I would say one aspect is interpretability.

So if you build explicit forward models and someone communicate the belief that the agent holds, that is, you know, the trigger for the agent taking certain actions, then that could help explain and thereby build trust, right?

So this causal chain of internal belief to agent action is harder to disentangle if you look at things like large language models.

Obviously, it now comes in where, you know, they make more explicit their sort of chain of thoughts.

But I would say that there is one, maybe one aspect that can help with user trust.

So there are the, there's elements of it.

So there's some bits that would be on the design side.

So if we have explicit models and we can look at a particular interaction context and analyze it, then we can say, oh, okay, we understand why the system was behaving the way it was.

And we can decide whether that's sort of good or bad thing.

We may still need to create a succinct metaphor for interaction for the user where the predictions are fed in and that can communicate that succinctly to the user.

So in some settings, if this was a, you know, so say you were controlling a tanker and an active inference navigation system was giving you, it could be saying my predictions about where you're going to be if you do this thing or this, and then you could relate to that.

Another way it can potentially build trust is because you're closing loops to, you know, maximize your predictability and minimizing your surprise.

It can potentially, there might be two possible interpretations of the user situation and it could try and do something to quickly reduce the uncertainty of which it's in and then have a more coherent, feedback for them.

feedback for them.

Whereas in other situations, the system would have to say, well, if, if this was your intention, then this would

be happening if this was you.

So an active inference system could try and reduce that uncertainty more rapidly in order to give a more coherent feedback and control.

So that might be one way to build up trust that you can see, okay, it thinks I'm doing this.

Again, it's about the mutual modeling aspect that we were talking about.

As the user starts to become more confident that they understand the system, it can work with that.

So if, if it understands that they don't yet have, the user doesn't seem to have a good model of them.

So if the active inference agent has an explicit model of the user's model of that, that model is still quite uncertain.

It knows there's not an awful lot of points in having a very deep prediction horizon because very soon the user is going to get so diffuse about their understanding of what's going on.

So it maybe needs to spend more effort in the early stages of improving the user's model of it.

So it's maybe has to perform some tasks, not so much to achieve the user's goals, but to help to reduce the uncertainty of the user's model of it.

So there may be a stage where the system just automatically essentially trains the user how capable it is.

And it's also then improving its model of the user and vice versa.

So you may find that it achieves trust by going through a process of reducing the uncertainty of the mutual models, and then they can do more together.

The prediction horizons gradually get longer because they're able to predict each other's behavior more coherently.

Yeah, being able to explicitly disentangle pragmatic and epistemic value.

And when are we seeking to provide preferred sensations like staying on the road for the self-driving car or the songs that are preferred for a given listener?

And then epistemic value can provide learning and updating.

It's not needed in every single situation.

Just taking ad hoc excursions from blood sugar homeostasis is probably deleterious.

But then again, it would be in balance with maintaining it in a healthy range to understand responsivity.

So it's like whether for biological systems or these more synthetic informational systems, being able to understand and have an explicit dial.

And this is also one of the most challenging parts of modeling, even after you've set up the pragmatic and epistemic value terms, their balance is like doing control theory on a seesaw because it contains a higher order trade-off with preference satisfaction and with learning.

Aaron?

Aaron?

Aaron?

Yeah, I just want to come in with, I think the RX and FUR team originally had an application where they were looking at hearing aids and parameter tuning and optimization specifically to help users use hearing aids more because the situations where you might be using that hearing aid would require very different parameters.

So if you're in a train station versus in your house, your hearing aid would need to be tuned differently, but

that's a very difficult procedure to do.

So they would have an active inference algorithm that would try and understand the uncertainty of where the user is, but deliberately choose odd parameter choices to learn about the context to then choose good parameter values in order that the user has a good hearing experience.

[illegible]

clear, explicit

modeling of uncertainty.

How do we communicate

that to users

in a sensible way? You also

mentioned the large language model stuff and Jane of

Thought. I think

they don't have

an explicit model of uncertainty,

but they do

a good job of communicating.

Whether what they're communicating is actually

valid or not is a separate question.

But they give

English language that people can understand, right?

There's a lot of work

on how to represent

distributions interactively.

You can have things like audio

models, which are

properties of the sound can represent

the distributions involved. You've got

visual models where you can blur

images. But I think

you're going to end up with

things like

Charnoff faces or

Charnoff faces.

you can remember the

Charnoff

radiation.

And Charnoff faces where you could

have multiple dimensions of a

face representing a

high dimensional data, but you could potentially

add extra properties to it.

So I think

there's going to be lots of scope for

creative

metaphor

design to represent

uncertainty.

creativity. We've been looking at this on and off over the years.

So

that's going to be something where

there's going to be loads of scope for

creativity. So

you could separate the

algorithmic elements of

estimating your beliefs from the way you

display

them to the user and then try and

optimize

which ways are working, how do you change it,

how do you, if you're not, if a particular approach is not

representing, is not communicating its uncertainty well to the user,

then we need to learn better ways of doing that. So there's going to be an element of

data driven tuning based on how well people are communicating and there's going to be an element of just creative

leaps of the imagination and how do we, you know, how do we take high dimensional uncertain data and represent them.

And just to go back, some of these things have been around for a long time. So, you know, the Bayesian

modelers at Microsoft had to think carefully about the wizards, you know, so if you're doing something mundane like a printer wizard,

there may be

an optimal question to ask next in terms of

expected

information gain.

but

it may be, it may, if

you ask a series of questions which seem to be about a topic, you know, are you properly plugged into the printer?

And then it suddenly leaps away and asks

something about a completely different topic and then comes back to

wires and printing and, you know, printer wires and so on.

The user may lose trust in it, even if it was a statistically it was an optimal question to ask

things. And because the user can't see the relationship and thinks this thing doesn't know it's just guessing random things.

So there may be an information theoretically optimal way to ask things.

But if you don't take into account the user's model, then, and so if the user is an experienced user and trusts the system, then that would be an optimal thing to do. If the user is a naive user in terms of this wizard, which is often the case because things should hopefully not fail all the time, then you have to assume that they are a bit mistrustful about products from Microsoft and

that they are willing to just quit using the wizard if it starts to ask what they think of as random questions. So this has been around for a long time. And it's going to be an interesting question. So one of the, you know, for example, in terms of the, the interaction between the

the human and the computer, the human and the human and the computer, the reason that we wanted to work on optimal interaction mechanisms as a topic is that we think there's going to be a very limited set of core features on the interaction, which can be combined in different ways with different sensors. And we want to have some sort of

framework that people can quickly recognize, and it's the way that nowadays toddlers pick up a magazine and start to swipe it thinking it's an iPad. They, they've very quickly grasped the basic ways of interacting with a tablet. And they're trying to generalize that around the world. And we think that, you know, a big part of making active inference and HCI work well together is going to be picking appropriate metaphors, building appropriate interaction mechanisms so that even if somebody is using a

set a new sensor for the first time, they can quickly manipulate it and see, okay, it seems to be mapping to a dial or a slider, and I'm hearing some formative feedback from the core mechanism that I'm familiar with. And I can quickly adjust my behaviour to work with the possible mappings that might be involved there. So it's going to, there's going to be a co creation of interaction metaphors, at the same time as the active inference and HCI come together.

And then all the authors, how are you planning to take the work forward?

And or what would be on your wish list?

Slash issues or questions to raise up for people to explore more?

Well, the work is continuing forward and we're, we're hiring exciting new PhD students, some of them were asking questions, which you weren't transmitting, but we're sure that we'll get a chance to hear that from them in person. So hopefully they'll be pushing things forward.

But I think the, this also, Aaron, you can see you've been voted in as a new speaker sometimes does that. So I think the questions that we have for the community would, you know, be about things like the preference priors, you know, how are people working with that and tuning that and eliciting that in other contexts?

And adapting them, learning them from behaviour, because that's all going to be super relevant for building systems that can interact well as a human and reasoning about, you know, what should, should the priors on the system just be about pleasing the human? Should we have other constraints on them?

What does that mean for something like an educational app, which should know more than the user about certain things at the start of the process? Because the whole point is to change the human.

So I think the, you know, we're, we're excited to hear what's happening on things like preference priors on the whole debate about alternatives to EFE.

And for prediction because what happens when things have not yet converged because some of the assumptions in the core cost functions like EFE are assuming that the system's already doing the right thing and what happens when there's a mismatch at the start of that process.

So I think they're getting more input from people who are experienced working with these things would be exciting for us.

Those of you have any, anything more you want to add in?

Yeah, I think we've still got some questions around how the learning of the models happens.

So how do you learn a power model, particularly how do you do that during interaction without introducing instability?

The question is about the representation of the low level distributions, the kind of things Sebastian was talking about.

You can't always work with uni model models, even want to do things in continuous spaces.

How can we do that for the kind of distributions we actually encounter in human computer interaction?

And questions about how we do the debugging of these kinds of models?

What are the diagnostics we can apply to see how well these models are working and track that?

We're going to build them and evaluate them.

Something which is going to be core to that.

I think for me, one of the most interesting things is looking at scaling so we can scale things up to higher dimensional user inputs, but also scaling to much higher dimensional degrees of freedom in terms of the system that's being controlled.

And the time horizon, so for which we reason and that then brings up questions about, you know, what do we amortize?

What do we actually explicitly model and roll out and whether or not we bring in some concepts of, you know, hierarchical agents and so on.

So I think scaling for me and everything around that sort of what's going to keep me busy for the next, next one.

I think just to just to add there as well, I think the scale free part is particularly, particularly a point of focus right now.

So we've had the scale for active inference paper that came out last year.

I think there's a paper that came out this year about the tree search and trying to reduce the sheer competition or explosion of policy policies that you can have if you have a planning horizon longer than X .

So I think we'll see a lot of work addressing those issues over the next few years.

For me, it's just a case of getting really clear and reproducible software packages so that we can make this easy.

I think that's, that's where, that's where the work will be over the next few years to just being able to say, right, create an agent, let it go.

The thought, the thinking should be about how you set up your generative model and not really about whether you're using effective free energy or expected free energy of the future or generalized free energy.

I think that's largely a problem that should not necessarily, that should not need to be solved by say designers or human computer interaction specialists.

I'd like it ideally that the software is, you know, sort of, batteries included off you go, you get active inference out of the box.

But then it's coming up with a model that maps to your problem, maps to the problem that you are trying to solve for a user.

So hopefully we get there sooner rather than later.

And I think also creating tools to, you know, harnesses to take these agents and their works and, you know, run simulations and analyze them in different contexts and visualize what's going on.

And look at counterfactual situations and so on.

So I think if this is, if we're doing it all right, then we have the tools to make the active inference work well.

But then we're trying to, you know, even if it's understanding a natural system, you might like to put some, put it into a particular context and observe what happens.

And doing the tracking of, you know, the behaviour, the kind of digital twin element of this.

So, how do we build active inference agents that we can actually monitor and analyze and say, okay, we're now understand, if we want to have interpretable systems, then we can get it into a particular context and statistically understand what's going on around it.

And why is, you know, is this down to, is the behaviour, is the lack of, you know, what seems to be a sensible action there down to an imperfect generative model of the world?

Is it down to too short a prediction horizon?

Is it that we're just aren't sensing the world well enough?

Then that's, that's just the limit of it's doing the right thing given the information it has.

But so I think creating the tooling to understand and analyze the models is going to be interesting as well.

So, what's the problem?

Well, you've created really a wonderful style sheet and almost a graphical taxonomy of different settings where that is coming into play in cyber physical systems, which have at least partially a fully modelable component.

And so that plus open source software development plus community feedback and application, it will be

exciting to see where we all go.

So thank you again to the authors and to everyone for raising the possibility of the stream.

So thanks for inviting us.

See you all.

Thank you.

Bye.

Bye.